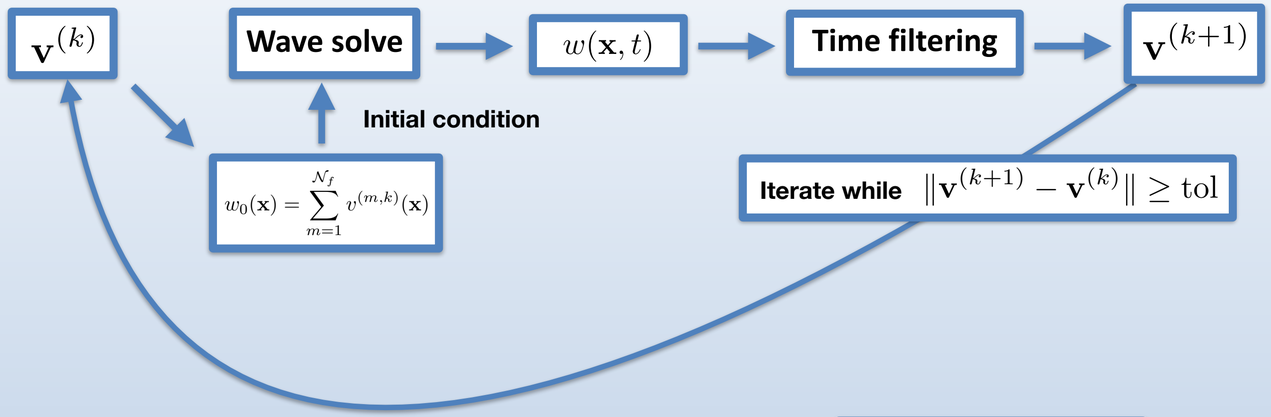
Computing multiple Helmholtz solutions by time filtering a single wave equation

Francis Appiah

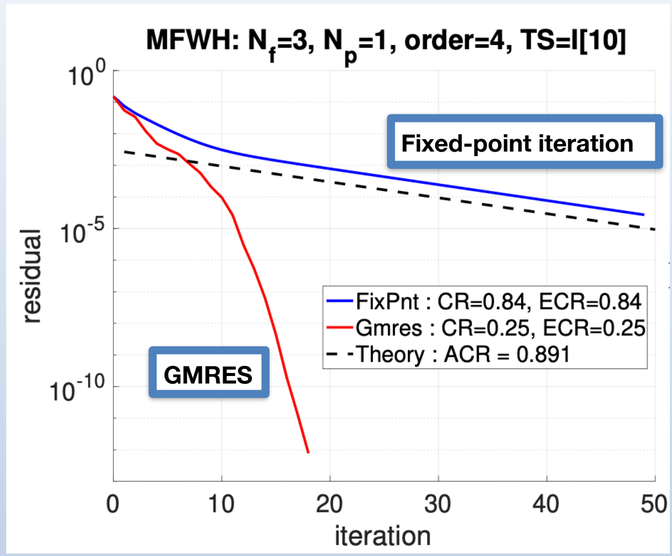
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MFWH: A Fix-point iteration that converges to multiple Helmholtz solutions simultaneously with different frequencies and/or forcing



At convergence: $v^{(m,k+1)} \to u^{(m)}, m = 1, 2, \ldots, \mathcal{N}_f$

No inversion of an indefinite Helmholtz matrix is required



Key Features

- Time domain wave solve
- Time filtering
- GMRES acceleration
- Optimal $\mathcal{O}(N)$
- Solutions on Overset grid